

T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2) (S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1) (S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2) (S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1 S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$16\pi^2\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$

Column Representative term structure	
T1	$\frac{55g_1^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{55g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$
Quantity: Y_u

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$-3\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$
T2	$+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$
T3	$+3\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$3\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	$+3\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

$\beta_{y_1}^{(1)}$
Quantity: y_1

$$16\pi^2\,{}_{\mu}\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$2T_{\nu}^{(1)}(y_1)_{ij}$
T2	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$
T3	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$2T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$

Quantity: y_1

Terms 2 of 2

Column	Representative term structure
T1	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T2	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$
T3	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$

$\beta_{y_2}^{(1)}$
Quantity: y_2

$$16\pi^2\,\mu\,\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$T_{\nu}^{(2)}(y_2)_{ij}$
T2	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$
T3	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Quantity: y_2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{3}{2}g_1^2(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2(y_2)_{ij}$

$\beta_{M_F}^{(1)}$

Quantity: M_F

$16\pi^2\,\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\,(y_1)^*\,M_F\right)_{ij}+3\left(M_F\,(y_2)^\dagger\,y_2\right)_{ij})$
T2	$-\frac{9}{2}g_2^{\,2}\,(M_F)_{ij}$
T3	$-\frac{3}{2}g_1^{\,2}\,(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\,(y_1)^*\,M_F\right)_{ij}+3\left(M_F\,(y_2)^\dagger\,y_2\right)_{ij})-\frac{9}{2}g_2^{\,2}\,(M_F)_{ij}$	$-\frac{3}{2}g_1^{\,2}\,(M_F)_{ij}$	

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\,\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$
T2	$-8\,\mathrm{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$
T3	$-6g_1^2m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\mathrm{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}m_2^2$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$
T2	$-4\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$
T3	$-12g_2^2m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}m_1^2$

$\beta_{\lambda_1}^{(1)}$

Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$
T3	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$
T2	$+4\lambda_5\,(\lambda_5)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$	$+4\lambda_5\,(\lambda_5)^*$	$+ \frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Terms 4 of 5

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+2\lambda_{HS_1}^{(1)2}$
T3	$+6\lambda_{HS_2}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)^2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$8\lambda_{S_1}^{(1)}T_{\nu}^{(1)}$
T2	$-8\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_1\,(y_1)^{\dagger}\Big)$
T3	$+12g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$8\lambda_{S_1}^{(1)}T_{\nu}^{(1)}$	$-8\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_1\,(y_1)^{\dagger}\Big)$	$+12g_1^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$-12g_1^2\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)2}$
T3	$+4\lambda_{HS_1}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+10\lambda_{S_1}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+10\lambda_{S_1}^{(1)2}$

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$
T2	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$
T3	$-3\text{Tr}\Big(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\Big(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+21g_2^4$
T2	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3	$+2\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$
T3	$+14\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+26\lambda_{S_2}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_2}^{(A)2}$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3	$+12\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T2	$+3g_1^4$
T3	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_1}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1) \ 2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu}=\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda^{(A)}_{S_2}\big)$
T2	$-2\operatorname{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda^{(A)}_{S_2}\big)$	$-2\operatorname{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T3	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4\lambda_{HS_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1) \ 2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+6\lambda_{HS_2}^{(A) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$
T2	$-4\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\Big)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\Big)$	$+4\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$-12g_2^2\lambda_{12}^{(1)}$
T2	$-6g_1^2\lambda_{12}^{(1)}$
T3	$+4\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_5}^{(1)}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$
T2	$-\frac{21g_2^2\lambda_5}{2}$
T3	$-\frac{9g_1^2\lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$
T2	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$
T2	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

$$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{S_2}^{(A)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$
T2	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$
T3	$-\text{Tr}\Big(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+ \frac{3 g_2^4}{4}$
T2	$+ g_2^2 (-3 \lambda_{S_2}^{(1)} - 24 \lambda_{S_2}^{(A)})$
T3	$+ \frac{\lambda_{12}^{(1)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+ \frac{3 g_2^4}{4}$	$+ g_2^2 (-3 \lambda_{S_2}^{(1)} - 24 \lambda_{S_2}^{(A)})$	$+ \frac{\lambda_{12}^{(1)2}}{4}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$
T2	$+3\lambda_{HS_2}^{(A) \ 2}$
T3	$+\frac{7\lambda_{S_2}^{(1) \ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) \ 2}}{4}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+22\lambda_{S_2}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+22\lambda_{S_2}^{(A)^2}$